Practice assignment 8 - Not Graded

This assignment will not be graded and is only for practice.
Note:Note: Suppose W_1 and W_2 W1 and W2 are subspaces of a vector space V V . Then sum of two subspaces is defined as:

$$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\}.$$ W1 + W2 = {w1 + w2 | w1 ∈ W1, w2 ∈ W2}.

One can verify that $W_1 + W_2$ W1 + W2 is a subspace of V V .

Multiple Choice Questions (MCQ):

1 point
An inner product on R^3 R3 is defined as:

$$\langle .,. \rangle : R^3 \times R^3 \to R$$ ⟨., .⟩: R3 × R3 → R

$$\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1y_1 + x_2y_2 + x_3y_3$$ ⟨(x1, x2, x3), (y1, y2, y3)⟩ = x1y1 + x2y2 + x3y3.

Match a set S S in Column A with the set T T given in column B, so that $T \subset Span(S)^{\perp}$ T ⊂ Span(S)⊥. Match a set in column B with the property it satisfies in column C.

Which of the following option are true?

- [ ] d → → ii → → 3
- [ ] d → → iii → → 4
- [ ] c → → ii → → 4
- [ ] a → → iv → → 1
- [ ] a → → ii → → 3
- [ ] b → → iv → → 2

No, the answer is incorrect.
Score: 0
Accepted Answers:
d → → iii → → 4
a → → ii → → 3

Suppose W_1 and W_2 W1 and W2 are subspaces of a vector space V V . Let P_{W_1} and P_{W_2} PW1 and PW2 denote the projection from V V to W_1 W1 and V V to W_2 W2 respectively and consider the following statements:

- **Statement P**: If $P_{W_1} + P_{W_2}$ PW1 + PW2 is a projection from V V to $W_1 + W_2$ W1 + W2, then $P_{W_1} \circ P_{W_2} + P_{W_2} \circ P_{W_1} = 0$.PW1 ∘ PW2 + PW2 ∘ PW1 = 0.

- **Statement Q**: $P^2_{W_1} := P_{W_1} \circ P_{W_1}$ PW1 2 := PW1 ∘ PW1 is not a projection from V V to W_1 W1.

- **Statement R:** If A is the matrix representation of P_{W_2}, then A can not be a symmetric matrix.
- **Statement S**: $P_{W_1} - P_{W_2}$ is a projection from V to $W_1 + W_2$.

Find the number of correct statements

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

1 point
Which of the followings option(s) is (are) true?

- [ ] Let $u, v,$ and w be three vectors in R^2 and $\langle ., . \rangle$ be an inner product on R^2. If $\langle u, v \rangle = \langle u, w \rangle$, then $v = w$
- [ ] Let V be an inner product space. If $\langle u, v \rangle = 0$ for all $u \in V$, then $v = 0$
- [ ] Let V be an inner product space and $u, v \in V$. If $\langle u+v, u-v \rangle = 0$, then $\|u\| \neq \|v\|$
- [ ] Let V be an inner product space. If $S \subset V$ then $S^{\perp} = (span(S))^{\perp}$
- [ ] Suppose $\{u_1, u_2, \ldots u_m\}$ be an orthogonal set of vectors in an inner product space V, then $\|u_1 + u_2 + \ldots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \ldots + \|u_m\|^2$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Let V be an inner product space. If $\langle u, v \rangle = 0$ for all $u \in V$, then $v = 0$
Let V be an inner product space. If $S \subset V$ then $S^{\perp} = (span(S))^{\perp}$
Suppose $\{u_1, u_2, \ldots u_m\}$ be an orthogonal set of vectors in an inner product space V, then $\|u_1 + u_2 + \ldots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \ldots + \|u_m\|^2$.

1 point
Which of the following option(s) is/are true?

- [ ] Let A and B be orthogonal matrices of order 3. Then A and B are equivalent matrices.
- [ ] Let A and B be orthogonal matrices. Then A and B are similar matrices.
- [ ] Let A be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θ. Then nullity of the matrix A is 0.
- [ ] Let A be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θ and B be a rotation matrix corresponding to the anti- clock wise rotation of the YZ-plane about the X -axis with angle β. Then A and B are equivalent matrices.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Let A and B be orthogonal matrices of order 3. Then A and B are equivalent matrices.
Let A be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θ. Then nullity of the matrix A is 0.
Let A be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θ and B be a rotation matrix corresponding to the anti- clock wise rotation of the YZ-plane about the X -axis with angle β. Then A and B are equivalent matrices.

Numerical Answer Type:

Let $u = (1, 2, 1)$ and $v = (1, 2, 3)$ be the vectors of the inner product space $\mathbb{R}^3$ with usual inner product. Let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be an orthogonal linear transformation. If θ is an angle between $T(u)$ and $T(v)$, then find the value of $||T(u)||\,||T(v)||\cos\theta$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 8
1 point
Consider a set $W = \{(1, 1, 1)\}$ from the inner product space $\mathbb{R}^3$. Find the dimension of the subspace $W^{\perp}$, where $W^{\perp}$ is the collection of the vectors which are orthogonal to the vector (1,1,1).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
Let $v = (3, 1, 2)$ be a vector in $\mathbb{R}^3$. If (a, b, c) is the vector obtained from v after the anti- clock wise rotation of ZX-plane with angle 45° about the Y- axis, then find the value of $\sqrt{2}(a + b + c - 1)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 6
1 point

Comprehension Type Question:

With a particular frame of reference (in $\mathbb{R}^2$) the position of Rahul's house is at the point (8,1). There are two roads Road 1 and Road 2 along the lines $y = 2x$ and $y = -2x$ respectively on the $XY-$plane.

Answer questions 8,9 and 10 using the given information.

1 point

Which of the following option(s) is (are) true?

- [ ] If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).
- [ ] If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at $\frac{6}{5}(1, -2)$56(1,−2).
- [ ] If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at $\frac{6}{5}(1, -2)$56(1,−2).
- [ ] If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at (2,4).

No, the answer is incorrect.

Score: 0

Accepted Answers:

If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).

If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at $\frac{6}{5}(1, -2)$56(1,−2).

Let θθ be the acute angle between Road 1 and Road 2, then what will be the value of $5\cos(\theta)$5 cos(θ)?

No, the answer is incorrect.

Score: 0